Exercise Sheet

Lecture 10

The Spectral Theorem

The sheet uses finite-dimensional real or complex inner-product spaces; quantum measurement questions are in the finite ideal-projective model. Inner products are in the physics convention; the adjoint satisfies $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. A ★ marks a harder problem.

Core tutorial route

Work **10.4, 10.7, 10.8, 10.10, 10.16, 10.18, 10.22, and 10.25** in that order (about 80–100 minutes before discussion). This eight-problem route covers complex normal/Hermitian diagonalization, completion of the normal-triangular proof, the finite self-adjoint criterion, positive functional calculus, spectral projector interpolation, simultaneous commuting self-adjoint reasoning, a degenerate measurement transfer, and probability normalization.

Additional practice: 10.1, 10.3, 10.6, 10.9, 10.11–10.15, 10.17, 10.19–10.21, 10.23–10.24. **Bridge:** 10.2 and 10.5 (real-entry normal operators viewed over $\mathbb{C}$; contrast their lack of real orthogonal diagonalization). **Extension:** none. Bridge tasks do not count toward core progress.

Diagonalization in an orthonormal basis

Exercise 10.1 (computational) Diagonalize the Hermitian matrix $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ in an orthonormal basis: find its eigenvalues, an orthonormal eigenbasis, and the unitary U with $U^\dagger A U$ diagonal.

Answer. Eigenvalues $3, -1$; $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $U^\dagger A U = \operatorname{diag}(3, -1)$.

Exercise 10.2 (computational) On $\mathbb{C}^2$, show that $A = \left(\begin{smallmatrix} 0 & -2 \\ 2 & 0 \end{smallmatrix}\right)$ is normal but not self-adjoint, and find a unitary diagonalization. Explain why the same real matrix has no orthogonal diagonalization on $\mathbb{R}^2$.

Answer. Over $\mathbb{C}$: normal but not self-adjoint; eigenvalues $\pm 2i$ on $\frac{1}{\sqrt{2}}(1, \mp i)$. Over $\mathbb{R}$: no eigenvalues, hence no orthogonal diagonalization.

Exercise 10.3 (computational) Diagonalize the real symmetric matrix $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ in an orthonormal basis: give the eigenvalues, an orthonormal eigenbasis, and the orthogonal O with $O^\mathsf{T} A O$ diagonal.

Answer. Eigenvalues $4, 2$; $O = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $O^\mathsf{T} A O = \operatorname{diag}(4, 2)$.

Exercise 10.4 (computational) Treat the following as a complex normal operator, unitarily diagonalize it, and then identify the stronger property that forces its eigenvalues to be real: $A = \left(\begin{smallmatrix} 1 & 2i \\ -2i & 1 \end{smallmatrix}\right)$: find its (real) eigenvalues, an orthonormal eigenbasis, and the unitary U with $U^\dagger A U$ diagonal.

Answer. A is Hermitian (hence normal); eigenvalues $3, -1$ and $U = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} i & -i \\ 1 & 1 \end{smallmatrix}\right)$ gives $U^\dagger A U = \operatorname{diag}(3, -1)$.

Exercise 10.5 (computational) On $\mathbb{C}^2$, show that $A = \left(\begin{smallmatrix} 2 & -3 \\ 3 & 2 \end{smallmatrix}\right)$ is normal but not self-adjoint, and give a unitary diagonalization. Explain why there is no real orthogonal diagonalization on $\mathbb{R}^2$.

Answer. Over $\mathbb{C}$: normal, not self-adjoint, with eigenvalues $2 \pm 3i$ on $\frac{1}{\sqrt{2}}(1, \mp i)$. Over $\mathbb{R}$: no eigenvalues, hence no orthogonal diagonalization.

Exercise 10.6 (computational) Diagonalize the real symmetric $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{smallmatrix}\right)$ on $\mathbb{R}^3$ in an orthonormal basis, and give $O^\mathsf{T} A O$.

Answer. Eigenvalues $2, 2, 4$ on $(1, 0, 0)$, $\frac{1}{\sqrt{2}}(0, 1, -1)$, $\frac{1}{\sqrt{2}}(0, 1, 1)$; $O^\mathsf{T} A O = \mathrm{diag}(2, 2, 4)$.

Exercise 10.7 (proof) In the proof of the finite complex spectral theorem, T is normal and upper-triangular in an orthonormal basis $e_1, \ldots, e_n$. Show the first-row off-diagonal entries vanish, identify $A = a_{11} \oplus A_1$, prove the trailing block A_1 is normal, and complete the induction that A is diagonal.

Exercise 10.8 (proof) Let T be normal on a finite-dimensional complex inner-product space. Using the spectral theorem, prove that T is self-adjoint if and only if every eigenvalue of T is real.

Spectral decomposition

Exercise 10.9 (computational) Write the spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2$ of $\sigma_z = \mathrm{diag}(1, -1)$, and verify $P_1 + P_2 = I$ and $P_1 P_2 = 0$.

Answer. $\sigma_z = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$; $P_1 + P_2 = I$, $P_1 P_2 = 0$.

Exercise 10.10 (computational) Using the spectral form, compute $f(A)$ for $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ and $f(\lambda) = \lambda^{1/2}$.

Answer. $\sqrt{A} = \frac{\sqrt{3}+1}{2} I + \frac{\sqrt{3}-1}{2} \sigma_x$.

Exercise 10.11 (computational) Write the spectral decomposition $A = \lambda_1 P_1 + \lambda_2 P_2$ of $A = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 4 \end{smallmatrix}\right)$, giving the projectors explicitly, and verify $P_1 + P_2 = I$ and $P_1 P_2 = 0$.

Answer. $A = 6P_+ + 2P_-$ with $P_\pm = \frac{1}{2}\left(\begin{smallmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{smallmatrix}\right)$.

Exercise 10.12 (computational) Using the spectral form, compute the inverse A^{-1} of $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.

Answer. $A^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$.

Exercise 10.13 (computational) Find the positive square root $\sqrt{A}$ of the positive operator $A = \left(\begin{smallmatrix} 10 & 6 \\ 6 & 10 \end{smallmatrix}\right)$ via the spectral form.

Answer. $\sqrt{A} = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$.

Exercise 10.14 (computational) Reconstruct the self-adjoint operator A on $\mathbb{C}^2$ with eigenvalue 3 on $\frac{1}{\sqrt{2}}(1, 1)$ and eigenvalue 7 on $\frac{1}{\sqrt{2}}(1, -1)$.

Answer. $A = \left(\begin{smallmatrix} 5 & -2 \\ -2 & 5 \end{smallmatrix}\right)$.

Exercise 10.15 (proof) Prove that an orthogonal projection P is self-adjoint and idempotent ($P^\dagger = P$, $P^2 = P$), and conversely.

Exercise 10.16 (proof) Let T be normal with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ and spectral projections P_k. Prove the interpolation formula $P_k = \prod_{j \neq k} \dfrac{T - \lambda_j I}{\lambda_k - \lambda_j}$.

Measurement and commuting operators

Exercise 10.17 (computational) Measure σ_z on $|\psi\rangle = \frac{1}{\sqrt{10}}(|0\rangle + 3|1\rangle)$. Find the probabilities of ± 1 and the expectation $\langle\sigma_z\rangle$.

Answer. $P(+1) = \frac{1}{10}$, $P(-1) = \frac{9}{10}$; $\langle\sigma_z\rangle = -\frac{4}{5}$.

Exercise 10.18 ($\star$) Let V be finite-dimensional. If $A, B \in \mathcal{L}(V)$ are commuting self-adjoint operators, prove that they have a common orthonormal eigenbasis.

Hint. B sends each eigenspace of A into itself (since $A(Bv) = B(Av)$), so apply the spectral theorem to B restricted to each one.

Exercise 10.19 (computational) On $\mathbb{C}^3$, let $A = \operatorname{diag}(1, 1, 4)$. Find the orthogonal projection onto the 1-eigenspace and the probability that measuring A on $|\psi\rangle = \frac{1}{\sqrt{3}}(1, 1, 1)$ yields 1.

Answer. $P_1 = \operatorname{diag}(1, 1, 0)$; probability of outcome 1 is $\frac{2}{3}$.

Exercise 10.20 (computational) Measure σ_z on $|\psi\rangle = \frac{1}{2}(\sqrt{3}\,|0\rangle + |1\rangle)$. Find the probabilities of ± 1 and the expectation $\langle\sigma_z\rangle$.

Answer. $P(+1) = \frac{3}{4}$, $P(-1) = \frac{1}{4}$; $\langle\sigma_z\rangle = \frac{1}{2}$.

Exercise 10.21 (computational) Measure $A = \operatorname{diag}(1, 2, 3)$ on $|\psi\rangle = \frac{1}{\sqrt{6}}(1, 1, 2)$. Find the probabilities of the three outcomes and $\langle A\rangle$.

Answer. $P(1) = \frac{1}{6}$, $P(2) = \frac{1}{6}$, $P(3) = \frac{2}{3}$; $\langle A\rangle = \frac{5}{2}$.

Exercise 10.22 (computational) On $\mathbb{C}^3$ let $A = \operatorname{diag}(2, 2, 5)$. Measure A on $|\psi\rangle = \frac{1}{3}(2, 1, 2)$: find the probabilities of each outcome, $\langle A\rangle$, and the post-measurement state when the outcome is 2.

Answer. $P(2) = \frac{5}{9}$, $P(5) = \frac{4}{9}$; $\langle A\rangle = \frac{10}{3}$; collapses to $\frac{1}{\sqrt{5}}(2, 1, 0)$.

Exercise 10.23 (computational) Verify that $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 4 \end{smallmatrix}\right)$ commute, and simultaneously diagonalize them in one orthonormal basis.

Answer. $AB = BA = \left(\begin{smallmatrix} 6 & 9 \\ 9 & 6 \end{smallmatrix}\right)$; shared basis $\frac{1}{\sqrt{2}}(1, \pm 1)$, with $A = \operatorname{diag}(3, -1)$, $B = \operatorname{diag}(5, 3)$.

Exercise 10.24 ($\star$) Suppose T is normal with $T^3 = T$. Show that the eigenvalues of T lie in $\{0, 1, -1\}$ and that T is self-adjoint.

Hint. Use the spectral decomposition $T = \sum_k \lambda_k P_k$, so $T^3 = \sum_k \lambda_k^3 P_k$; compare eigenvalue by eigenvalue.

Exercise 10.25 (proof) For an observable $A = \sum_k \lambda_k P_k$ with the spectral projections P_k, prove that the Born-rule probabilities $\|P_k|\psi\rangle\|^2$ of a normalized state $|\psi\rangle$ sum to 1.